

Physics Homework

Uniform Circular Motion & Universal Gravitation

Name: _____

Date: _____

Constants:

$$g = 9.8 \text{ m/s}^2$$

$$G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2,$$

$$M_E = 5.97 \times 10^{24} \text{ kg}$$

$$R_E = 6.37 \times 10^6 \text{ m}$$

Directions: Show all work. Include units in every step. Box or circle your final answers.

Part A: Uniform Circular Motion (Centripetal Motion)

A1. A 0.50 kg ball moves in a horizontal circle of radius 1.6 m at a constant speed of 4.0 m/s.

(a) Find the centripetal acceleration a_c .

(b) Find the centripetal force F_c on the ball.

(c) If the speed doubles, what happens to a_c ? (State the factor change and briefly explain.)

A2. A car of mass 1200 kg goes around a flat circular turn of radius 60 m. The coefficient of static friction between the tires and road is $\mu_s = 0.75$.

(a) What is the *maximum* speed the car can have without skidding?

(b) In this situation, what provides the centripetal force?

A3. A roller coaster cart has mass 900 kg and travels at 18 m/s through a dip that forms part of a vertical circle of radius 25 m.

(a) At the *bottom* of the dip, find the centripetal acceleration.

(b) At the bottom, is the normal force on the cart greater than, equal to, or less than its weight? Justify using a force equation.

A4. A satellite must orbit Earth in a circle of radius 7.0×10^6 m (measured from Earth's center) with speed 7.5×10^3 m/s.

(a) Find the centripetal acceleration required for this orbit.

(b) Find the orbital period T .

Part B: Universal Law of Gravitation

B1. Two objects have masses $m_1 = 5.0$ kg and $m_2 = 12.0$ kg and their centers are separated by 0.80 m.

(a) Calculate the gravitational force between them.

(b) If the distance between them is tripled, what happens to the force? (State the factor change.)

B2. A 55 kg person stands on Earth's surface.

(a) Using $F_g = \frac{GM_E m}{(R_E)^2}$, calculate the gravitational force on the person.

(b) Compare your answer to mg . Are they close? Briefly explain why.

B3. A spacecraft of mass 1500 kg is at an altitude of 400 km above Earth's surface.

(a) Find the distance from Earth's center to the spacecraft.

(b) Calculate the gravitational force on the spacecraft at this altitude.

(c) Calculate the gravitational acceleration g' at this altitude.

B4. The Moon has mass $M_m = 7.35 \times 10^{22}$ kg and the average Earth–Moon center-to-center distance is $r = 3.84 \times 10^8$ m.

(a) Calculate the gravitational force between Earth and the Moon.

(b) Using Newton's 3rd law, state the force the Moon exerts on Earth (magnitude and direction).

Part C: Combining Gravitation & Circular Motion

C1. A satellite orbits Earth in a circular orbit at altitude 300 km.

- (a) Find the orbital radius r from Earth's center.
- (b) Set gravitational force equal to centripetal force and solve for the orbital speed v .
- (c) Find the orbital period T .

C2. A planet has mass $M = 6.4 \times 10^{23}$ kg. A small moon orbits it in a circle of radius 2.0×10^7 m with period 3.0×10^4 s.

- (a) Find the moon's orbital speed using $v = \frac{2\pi r}{T}$.
- (b) Use $F_g = F_c$ to determine the mass of the moon (show your setup). *Hint:* you may find that the moon's mass cancels.

C3. Conceptual: A satellite is moved from orbit radius r to orbit radius $2r$ around the same planet.

- (a) By what factor does the orbital speed change?
- (b) By what factor does the orbital period change?